Zayaan Bhanwadia

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Education

University of Toronto Sept. 2025 – June 2028

HBSc in Computer Science – Software Engineering Specialist, Statistics Minor (Co-op) | GPA: 3.7/4.0

- **Activities & Awards:** Dean's List, 3x Hackathon Winner, UTRA Web Developer, VP Technology @ CS Association
- **Coursework:** Software Design, Systems Programming, Data Structures & Algorithms, Computational Theory

Experience

Software Developer Intern *(Full Stack)* Sept. 2026 – Present

FreshBooks *Toronto, ON*

- Shortened bank page load time by **86% (700ms to 100ms)** by moving a blocking API request to the background.
- Cut latency by **92%** over **20 benchmark calls** by caching invoices and paginated data in Redis with TTL and LRU.
- Enabled concurrent bank imports by replacing **2 system-level locks** with row-level locks on IDs and duplicates.
- Reduced invoice email latency by **1–3s** by decoupling PDF generation across services with GCS event delivery.
- Unified Plaid and Yodlee bank connections across backend services and Ember for staged rollout to **100%** of users.

Autonomy Software Engineer *(Mapping & Planning)* July 2026 – Present

University of Toronto Formula Racing – Driverless *Toronto, ON*

- Developed mapping and path-planning software in C++ and ROS 2 for **Canada's first driverless FSAE car**.
- Replaced FastSLAM with EKF-SLAM, reducing localization compute time by **30%** while maintaining accuracy.
- Built ego-aligned track corridors from quintic splines using arc-length sampling across a configurable **15m** horizon.
- Expanded simulation data collection and monitoring by integrating **8 sensors**, including IMU, wheel-speed, and GPS.

Software Engineer Intern *(Backend AI)* July 2026 – Sept. 2026

FlyRank AI *Toronto, ON*

- Prevented duplicate charges under **50+** simultaneous requests by deduplicating payment events in the database.
- Built a centralized embeddable widget platform, routing form submissions to a dashboard via a **one-line** script tag.
- Secured a public endpoint with validation, rate limiting, and geolocation checks, cutting spam by **80%**.

Software Developer Intern *(Full Stack)* May 2026 – July 2026

KorraNet Creative *Remote*

- Optimized Meta OAuth token handling and storage, reducing token refresh failures by **40%+** for AI integrations.
- Reduced system crashes and failed API calls by **85%+** by adding error handling and retries for failed requests.

Projects

Embedded Key-Value Store | *C++, CMake, Multithreading, Linux*

- Built an embedded database with **16-way** sharded locking, boosting read throughput **5x (8M to 40M ops/sec)**.
- Implemented reader-writer locks so reads proceed during disk writes, preventing race conditions across **16 threads**.
- Built background compaction to purge stale and deleted keys, cutting disk usage by **45%** without pausing operations.
- Implemented crash recovery by persisting every update to a checksummed write-ahead log before applying changes.

Context Sync Extension | *TypeScript, VS Code Extension API*

- Developed a VS Code extension with **750+ downloads** that automates AI chat context sharing across environments.
- Reduced token use by **25%+** by modelling chats as a weighted graph and selecting context via shortest-path search.
- Compacted chat summaries by **30%+** by designing a Markdown schema to pack more context into fewer tokens.

Technical Skills

Languages: Python, Java, JavaScript, C++, C, TypeScript, SQL, Bash, HTML/CSS

Libraries/Frameworks: FastAPI, Django, Express.js, Node.js, React.js, Ember.js, SQLAlchemy, ROS 2

Developer Tools: Linux, Docker, Git, GitHub, Claude Code, Codex, Jira, Confluence

Cloud/Databases: PostgreSQL, MySQL, Redis, AWS, Azure, GCP, Vercel, Supabase, MongoDB